

Model Question Paper-I with effect from 2025

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1BMATE201

Second Semester B.E./B.Tech. Degree Examination Calculus, Laplace Transforms and Numerical Techniques

TIME: 03Hours

Max.Marks:100

- Note: 1. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**
 2. VTU Formula Hand Book is Permitted
 3. M: Marks, L: Bloom's level, C: Course outcomes

Module-1			M	L	C
Q 1.	a	Evaluate $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x+y+z) dx dy dz$.	6	L2	1
	b	Evaluate $\int_0^a \int_0^{\sqrt{a^2-y^2}} y\sqrt{x^2+y^2} dx dy$ by changing into polar co-ordinates.	7	L2	1
	c	With usual notations derive the relation between beta and gamma function $\beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$	7	L2	1
OR					
Q 2.	a	Evaluate $\int_0^3 \int_1^{\sqrt{4-y}} \frac{dy dx}{\log y}$ by changing the order of integration.	6	L2	1
	b	Find the area between parabolas $y^2 = 4ax$ and $x^2 = 4ay$ using double integration.	7	L2	1
	c	Prove that $\int_0^{\pi/2} \frac{d\theta}{\sin \theta} \times \int_0^{\pi/2} \sin \theta d\theta = \pi$	7	L2	1
Module-2					
Q 3.	a	Find the directional derivative of $f(x,y,z) = xy^3 + yz^3$ at the point $(2, -1, 1)$ in the direction of the vector $\vec{i} + \vec{j} + \vec{k}$.	6	L2	1
	b	Show that $\vec{F} = (2xy^2 + yz)\vec{i} + (2x^2y + xz + 2yz^2)\vec{j} + (2y^2z + xy)\vec{k}$ is a conservative force field (Irrotational). Find its scalar function ϕ such that $\vec{F} = \nabla\phi$.	7	L2	1
	c	Applying Green's theorem, evaluate $\int_C [(y - \sin x)dx + \cos x dy]$ where C is the plane triangle enclosed by the lines $y = 0$, $x = \frac{\pi}{2}$ and $y = \frac{2}{\pi}x$	7	L2	1
OR					
Q 4.	a	Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$, $x^2 + y^2 - z = 3$ at the point $(2, -1, 2)$.	6	L2	1
	b	Show that $\vec{F} = \frac{x\vec{i} + y\vec{j}}{x^2 + y^2}$ is both Solenoidal and Irrotational.	7	L2	1
	c	Applying Stoke's theorem, evaluate $\int_C [(x+y)dx + (2x-z)dy + (y+z)dz]$ where C is the boundary of the triangle with vertices $(2,0,0)$, $(0,3,0)$, $(0,0,6)$.	7	L2	1
Module-3					
Q 5.	a	Find the approximate root of the equation $\tan x + \tanh x = 0$ in $(2,3)$ by applying Regula-Falsi method.	6	L2	2

	b	Apply Newton divided difference formula to find f(4) given the data. <table><tr><td>x</td><td>0</td><td>2</td><td>3</td><td>6</td></tr><tr><td>f(x)</td><td>-4</td><td>2</td><td>14</td><td>158</td></tr></table>	x	0	2	3	6	f(x)	-4	2	14	158	7	L2	2
x	0	2	3	6											
f(x)	-4	2	14	158											
	c	Evaluate $\int_4^{5.2} \log x \, dx$ using Simpson's $\frac{3}{8}$ th & $\frac{1}{3}$ rd rule by taking 6 equal strips.	7	L2	2										
OR															
Q 6.	a	Find an approximate root of the equation, $x \log_{10} x = 1.2$ near 2.5 by applying Newton Raphson, method	6	L2	2										
	b	Use appropriate Newton's interpolation formula to fit a polynomial of degree three which takes the following values <table><tr><td>x</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>y</td><td>6</td><td>24</td><td>60</td><td>120</td></tr></table>	x	3	4	5	6	y	6	24	60	120	7	L3	2
	x	3	4	5	6										
y	6	24	60	120											
c	By applying Simpson's $\frac{1}{3}$ rd rule, evaluate $\int_0^5 \frac{dx}{4x+5}$ with 11 ordinates and hence find log 5	7	L2	2											
Module-4															
Q 7.	a	Apply Taylor's series method solve $\frac{dy}{dx} = 3x + y^2$ with $y(0) = 1$ and hence find $y(0.1)$ and consider upto 4 th degree.	6	L2	2										
	b	Solve $\frac{dy}{dx} = \frac{y-x}{y+x}$, with $y(0) = 1$ and hence find $y(0.1)$ by taking one steps using Runge-Kutta method of fourth order.	7	L2	2										
	c	Apply Milne's method to solve $\frac{dy}{dx} = x - y^2$ with the following data $y(0) = 0$, $y(0.2) = 0.02$, $y(0.4) = 0.0795$, $y(0.6) = 0.1762$, compute y at $x=0.8$,	7	L2	2										
OR															
Q 8.	a	If $y' + y + 2x = 0$, $y(0) = -1$ then find $y(0.1)$ by using Taylor's series method. Consider upto third order derivative terms.	6	L2	2										
	b	Find $y(0.2)$ by using modified Euler's method. Given that $y' = x + y$, $y(0) = 1$. Take $h=0.1$ and carry out two modifications at each step.	7	L2	2										
	c	Apply Adam-Bashforth method to find $y(0.4)$ from $y' = \frac{xy}{2}$, $y(0) = 1$, $y(0.1) = 1.0025$, $y(0.2) = 1.0101$, $y(0.3) = 1.0228$.	7	L2	2										
Module-5															
Q 9.	a	Find $L(t^2 e^{-3t} \sin 2t)$	6	L2	3										
	b	A periodic function of period 2a is defined by $f(t) = \begin{cases} E & \text{for } 0 \leq t \leq a \\ -E & \text{for } a < t \leq 2a \end{cases}$ where E is a constant. Show that $L\{f(t)\} = \frac{E}{s} \tanh\left(\frac{as}{2}\right)$.	7	L2	3										
	c	Express the function $f(t) = \begin{cases} \pi - t & , 0 < t \leq \pi \\ \sin t & , t > \pi \end{cases}$ in terms of unit step function and hence find its Laplace transform.	7	L2	3										
OR															
Q 10.	a	Find the inverse Laplace transforms of $\frac{2s^2-6s+5}{s^3-6s^2+11s-6}$	6	L2	3										
	b	Find $L^{-1}\left\{\log \sqrt{\frac{s^2+a^2}{s^2+b^2}}\right\}$	7	L2	3										
	c	Solve $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = 12t^2 e^{-3t}$ subject to the conditions $y(0) = y'(0) = 0$ by using Laplace transforms.	7	L3	3										

CBCS SCHEME

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1BMATE201

Second Semester B.E./B.Tech. Degree Examination, June/July 2026 Calculus, Laplace Transform and Numerical Techniques : EEE Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.
3. VTU Formula Hand Book is permitted.*

Module - 1				M	L	C
Q.1	a.	Evaluate, $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x+y+z) dy dx dz$		6	L3	CO1
	b.	Evaluate, $\int_0^1 \int_x^{\sqrt{x}} xy dy dx$ by changing the order		7	L3	CO1
	c.	Prove that $\beta(m,n) = \frac{\gamma(m)\gamma(n)}{\gamma(m+n)}$		7	L2	CO1
OR						
Q.2	a.	Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing to polar coordinates.		6	L3	CO1
	b.	Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ by double integration.		7	L2	CO1
	c.	Prove that $\int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin\theta}} \times \int_0^{\pi/2} \sqrt{\sin\theta} d\theta = \pi$		7	L2	CO1
Module - 2						
Q.3	a.	Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ along $2i - j - 2k$		6	L2	CO1
	b.	Find the $\text{div}(\vec{F})$, $\text{curl}(\vec{F})$ where $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$		7	L2	CO1
	c.	Show that $\vec{F} = (y+z)i + (z+x)j + (x+y)k$ is irrotational. Also find a scalar function ϕ such that $\vec{F} = \nabla\phi$		7	L2	CO1
OR						
Q.4	a.	Find the angle between the surfaces $x^2 + y^2 - z^2 = 4$ and $z = x^2 + y^2 - 13$ at $(2, 1, 2)$		6	L2	CO1
	b.	If $\vec{F} = xyi + yzj + zxk$, evaluate $\int_c \vec{F} \cdot d\vec{r}$ where c is the curve represented by $x = t$, $y = t^2$, $z = t^3$, $-1 \leq t \leq 1$		7	L3	CO1

Q.4	c.	Employ Green's theorem in a plane to show that the area enclosed by a plane curve c is $\frac{1}{2} \oint xdy - ydx$ and hence find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	7	L2	CO1
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Module - 3

Q.5	a.	Find a real root of $xe^x - 2 = 0$ correct to three decimal places using Newton - Raphson method.	6	L2	CO2												
	b.	The area of a circle (A) corresponds to diameter D is given below. Find the area corresponding to diameter 105 using appropriate interpolation formula. <table border="1" data-bbox="491 748 1050 824"> <tr> <td>D</td><td>80</td><td>85</td><td>90</td><td>95</td><td>100</td></tr> <tr> <td>A</td><td>5026</td><td>5674</td><td>6362</td><td>7088</td><td>7854</td></tr> </table>	D	80	85	90	95	100	A	5026	5674	6362	7088	7854	7	L2	CO2
D	80	85	90	95	100												
A	5026	5674	6362	7088	7854												
	c.	Evaluate $\int_0^{0.3} (1 - 8x^3)^{\frac{3}{2}} dx$ by applying Simpson's $\frac{3}{8}$ rule with 7 ordinates.	7	L3	CO2												

OR

Q.6	a.	Compute the real root of $x \log_{10} x - 1.2 = 0$ by using Regula falsi method. Carry out three iterations.	6	L2	CO2										
	b.	Use Newton's divided difference formula, find $f(4)$ <table><tr><td>x</td><td>0</td><td>2</td><td>3</td><td>6</td></tr><tr><td>y</td><td>-4</td><td>2</td><td>14</td><td>158</td></tr></table>	x	0	2	3	6	y	-4	2	14	158	7	L2	CO2
x	0	2	3	6											
y	-4	2	14	158											
	c.	Use Lagrange's interpolation formula to find y at $x = 10$ <table><tr><td>x</td><td>5</td><td>6</td><td>9</td><td>11</td></tr><tr><td>y</td><td>12</td><td>13</td><td>14</td><td>16</td></tr></table>	x	5	6	9	11	y	12	13	14	16	7	L2	CO2
x	5	6	9	11											
y	12	13	14	16											

Module - 4

Q.7	a.	Use Taylor's series method to find y at $x = 0.1$, considering terms upto the third degree given that $\frac{dy}{dx} = x^2 + y^2$ and $y(0) = 1$	6	L2	CO2
	b.	Given $\frac{dy}{dx} = 3x + \frac{y}{2}$, $y(0) = 1$. Compute $y(0.2)$ by taking $h = 0.2$ using RK method of fourth order.	7	L2	CO2
	c.	Given that $\frac{dy}{dx} = x - y^2$, $y(0) = 0$, $y(0.2) = 0.02$, $y(0.4) = 0.0795$, $y(0.6) = 0.1762$. Compute y at $x = 0.8$ by applying Milne's method.	7	L2	CO2

OR

Q.8	a.	Using modified Euler's method find $y(0.1)$ given that $\frac{dy}{dx} = x^2 + y$, $y(0) = 1$ by taking $h = 0.05$. Perform two iterations in each step.	6	L2	CO2
	b.	Use fourth order RK method to solve $(x + y)\frac{dy}{dx} = 1$, $y(0.4) = 1$ at $x = 0.5$ correct to four decimal places.	7	L2	CO2
	c.	If $y' = 2e^x - y$, $y(0) = 2$, $y(0.1) = 2.010$, $y(0.2) = 2.040$, $y(0.3) = 2.090$ find $y(0.4)$ correct to four decimal places by using Adams – Bashforth predictor and corrector method. Use corrector formula twice.	7	L2	CO2

Module – 5

Q.9	a.	Find $L[te^{-2t}\sin 4t]$	6	L2	CO3
	b.	Find the LT of $f(t) = \begin{cases} E, & 0 < t < a/2 \\ -E, & \frac{a}{2} < t < a \end{cases}$ where $f(t+a) = f(t)$	7	L2	CO3
	c.	Express the following in terms of unit step function and hence find LT $f(t) = \begin{cases} 1, & 0 < t \leq 1 \\ t, & 1 < t \leq 2 \\ t^2, & t > 2 \end{cases}$	7	L2	CO3

OR

Q.10	a.	Find $L^{-1}\left[\frac{s+5}{s^2-6s+13}\right]$	6	L2	CO3
	b.	Solve $y''(t) + 3y'(t) + 2y(t) = 0$ with $y(0) = 1$, $y'(0) = 0$ using laplace transform.	7	L2	CO3
	c.	Express in terms of unit step function and find LT $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 4t, & 2 < t < 4 \\ 8, & t > 4 \end{cases}$	7	L2	CO3

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Second Semester B.E./B.Tech. Degree Examination, June/July 2026
Calculus, Laplace Transform and Numerical Techniques :
EEE Stream

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks, L: Bloom's level, C: Course outcomes.

3. VTU Formula Hand Book is permitted.

Module – 1				M	L	C											
Q.1	a.	Evaluate $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x+y+z) dy dx dz$	7	L1	CO1												
	b.	Evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$ by changing the order of integration.	7	L2	CO1												
	c.	Prove that $\beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$	6	L1	CO1												
OR																	
Q.2	a.	Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by change into polar coordinates.	7	L1	CO1												
	b.	Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ by double integration.	7	L2	CO1												
	c.	Evaluate $\int_0^\pi \sqrt{\tan \theta} d\theta$ by expressing in terms of gamma functions.	6	L2	CO1												
Module – 2																	
Q.3	a.	Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at $(2, -1, 2)$.	7	L2	CO1												
	b.	Find the directional derivative of $\phi = \frac{xz}{x^2+y^2}$ at $(1, -1, 1)$ in the direction of $\vec{A} = \hat{i} - 2\hat{j} + \hat{k}$.	7	L2	CO1												
	c.	Show that $\vec{F} = \frac{x\hat{i} + y\hat{j}}{x^2+y^2}$ is both Solenoidal and Irrotational.	6	L3	CO1												
OR																	
Q.4	a.	If $\vec{F} = xy\hat{i} + yz\hat{j} + zx\hat{k}$ evaluate $\int_C \vec{F} \cdot d\vec{r}$, where C is the curve represented by $x = t, y = t^2, z = t^3$, t lies between $-1 \leq t \leq 1$.	7	L2	CO1												
	b.	Apply Green's Theorem evaluate $\oint (x^2 + y^2) dx + 3x^2y dy$ where C is the circle $x^2 + y^2 = 4$ traced in the positive sign of the upper half of the circle.	7	L3	CO1												
	c.	Apply Stoke's Theorem evaluate $\int_C \vec{F} \cdot d\vec{r}$ for $\vec{F} = y\hat{i} + z\hat{j} + x\hat{k}$ where C is the boundary of the upper half the sphere $x^2 + y^2 + z^2 = 1$.	6	L3	CO1												
Module – 3																	
Q.5	a.	Find the real root of the equation $x \log_{10} x = 1.2$ lies between 2 and 3 by Regula –Falsi method (perform three iterations).	7	L2	CO2												
	b.	Apply Newton's divided difference formula find $f(9)$ from the data <table><tr><td>x</td><td>5</td><td>7</td><td>11</td><td>13</td><td>17</td></tr><tr><td>f(x)</td><td>150</td><td>392</td><td>1452</td><td>2366</td><td>5202</td></tr></table>	x	5	7	11	13	17	f(x)	150	392	1452	2366	5202	7	L2	CO2
x	5	7	11	13	17												
f(x)	150	392	1452	2366	5202												
	c.	Apply Simpsons three-eighth rule evaluate $\int_0^{0.3} \sqrt{1-8x^3} dx$ taking seven ordinates.	6	L2	CO2												

OR

Q.6	a.	Find the real root of the equation $\cos x - xe^x = 0$ using Newton Raphson method correct to three decimal places taking $x_0 = 0.5$.	7	L2	CO2
	b.	Apply Newton's Forward Interpolation Formula find the interpolating polynomial and hence find $f(0.5)$ given that $f(0) = 1$, $f(1) = 0$, $f(2) = 1$, $f(3) = 10$.	7	L2	CO2
	c.	Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by Trapezoidal Rule taking six equal parts.	6	L2	CO2

Module - 4

Q.7	a.	Apply Taylor Series Method find $y(0.2)$ given that $\frac{dy}{dx} = 2y + 3e^x$ and $y(0) = 0$.	7	L3	CO2
	b.	Compute y at $x = 1.2$ and $x = 1.4$ for $\frac{dy}{dx} = \log(x+y)$ given $y(1) = 2$ with $h = 0.2$ by Modified Buler's method.	7	L2	CO2
	c.	Apply Milne's predictor-corrector method find y when $x = 0.8$ given $\frac{dy}{dx} = x - y^2$ and $y(0) = 0$, $y(0.2) = 0.02$, $y(0.4) = 0.07$, $y(0.6) = 0.17$.	6	L3	CO2

OR

Q.8	a.	Apply Taylor series method find $y(0.1)$ given that $\frac{dy}{dx} = x - y^2$ and $y(0) = 1$.	7	L3	CO2
	b.	Apply Runge-Kutta method of order four find $y(0.1)$ given $\frac{dy}{dx} = \frac{y-x}{y+x}$ and $y(0) = 1$ with $h = 0.1$	7	L2	CO2
	c.	Compute $y(0.4)$ given $\frac{dy}{dx} = 2e^x - y$ and $y(0) = 2$, $y(0.1) = 2.01$, $y(0.2) = 2.04$ and $y(0.3) = 2.09$ by applying Adam's-Bashforth Method.	6	L3	CO2

Module - 5

Q.9	a.	Find the Laplace transform of (i) $e^{-3t}\sin^2 2t$ and (ii) $\frac{1-e^{at}}{t}$	7	L3	CO3
	b.	Find the Laplace transform of periodic function with period $2a$ defined by $f(t) = \begin{cases} 1 & \text{for } 0 < t < a \\ -1 & \text{for } a < t < 2a \end{cases}$	7	L3	CO3
	c.	Express in terms of unit step function and hence find the Laplace transform of $f(t) = \begin{cases} \cos t & \text{for } 0 < t \leq \pi \\ 1 & \text{for } \pi < t \leq 2\pi \\ \sin t & \text{for } t > 2\pi \end{cases}$	6	L3	CO3

OR

Q.10	a.	Find the inverse Laplace transform of $\frac{4s+5}{(s-1)^2(s+2)}$	7	L3	CO3
	b.	Apply Convolution Theorem, find the inverse Laplace transform of $\frac{1}{(s+1)(s^2+1)}$	7	L3	CO3
	c.	Solve by Laplace Transform Method $y'' - 2y' + y = e^{2t}$ with $y(0) = 0$, $y'(0) = 1$	6	L3	CO3
